UNIT 2 · WHAT'S WRONG AND WHAT TO MEASURE

What a Correct Theory Must Deliver

Session 8 · The checklist GE-LAV must satisfy

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

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What we'll cover today

1. **Pre-conditions**
   What we've established so far
2. **Theoretical requirements**
   Three properties any replacement must have
3. **Empirical requirements**
   Five empirical tests
4. **Practical requirements**
   Implementation feasibility
5. **How GE-LAV measures up**
   Preview of how it satisfies the list

Pre-conditions: where we stand

After 7 sessions, we have established:

- ✓ **DCF systematically misvalues private capital**
  Confirmed by secondary market evidence — $73M gap on a $103M asset (S1)
- ✓ **Three structural failures of DCF**
  Stochastic premium, Jensen bias, McKean-Vlasov externality (S2)
- ✓ **Liquidity illusion is the gap that GE-LAV must close**
  DCF marks vs. secondary clearing prices differ by 5-50% (S3)
- ✓ **OU process is the mathematically appropriate model for L(t)**
  Mean reversion + Brownian shocks → bounded variance (S4)
- ✓ **Premium has term structure π(L, T)**
  Two-dimensional dependence: state AND horizon (S5)
- ✓ **IRR is biased; PME is partial; both ignore liquidity**
  Why we need GE-LAV-consistent metrics (S6-7)

Theoretical requirements

REQ 1

General equilibrium

Must price in a market-clearing framework where supply equals demand for liquidity. Partial equilibrium analyses (LA-CAPM) are insufficient.

→ Session 17: GE Market Clearing

REQ 2

Collective dynamics

Must capture the McKean-Vlasov externality: individual decisions affect the collective state. Standard SDE models do not.

→ Session 21: MFG Equilibrium

REQ 3

Strict nesting of existing theories

Must reduce to DCF when premium is constant, to LAV when partial equilibrium. Should not invalidate existing tools, but extend them.

→ Section 4.5 of the book

Empirical requirements: five tests

Any candidate theory must pass these tests against historical data (2003–2024):

Test 1: Term structure recovery	Recover empirical regime-dependent $\pi(L, T)$ from secondary market data.
Test 2: Jensen bias calibration	Predicted Jensen bias = 0.8–3.6% should match observed return alpha across asset classes.
Test 3: Crisis amplification	Predict GFC-era valuation gap of 4.31× DCF vs. GE-LAV at L = -1.5.
Test 4: Welfare gap quantification	Predict ~$300B/yr aggregate welfare loss across the $13T market.
Test 5: Out-of-sample prediction	Calibrated GE-LAV must successfully predict 2020 COVID shock and 2022 rate shock magnitudes.

How GE-LAV measures up: a scorecard preview

Foreshadowing — by Session 24, you'll have computed all of these:

Requirement	GE-LAV result	Session covered
General equilibrium framework	✓ Yes	Session 17
McKean-Vlasov collective dynamics	✓ Yes	Session 21
Reduces to DCF when premium constant	✓ Yes (proved)	Section 4.5 (book)
Recovers empirical $\pi(L, T)$	✓ 0.94 R^2	Session 17
Predicts Jensen bias by asset class	✓ 0.8 – 3.6%	Session 24
Predicts GFC amplification	✓ 4.31×	Session 16
Quantifies welfare gap	✓ 2.3%/yr · $300B	Session 24
Out-of-sample (COVID, 2022)	✓ Within ±20%	Session 16 + 19

Session 8 summary

What we accomplished today

1. Unit 1 has built the diagnostic case: DCF fails, secondary market shows it, OU models L(t), term structure exists
2. Three theoretical requirements: GE, collective dynamics, strict nesting
3. Five empirical tests: term structure, Jensen bias, crisis amplification, welfare gap, out-of-sample
4. GE-LAV passes all eight by Session 24 — Unit 2 begins the construction

Next session

Session 9: Midterm review · Session 10: Midterm exam